MIRROR WORLD · TEAM PIGEONS

Locating a drone is hard.

A receiver hears a signal for a few seconds, a camera catches a few pixels, and the pieces don’t quite line up. Merging them takes a trained model.

And good, faithful training data barely exists.

What such a model learns from:

signal strength | signal type | timestamps | images | labelled boxes | **+ the truth behind every record**

OUR ANSWER

Simulate the world. Measure it. Export it.

Faithful by construction: every value comes from terrain and physics.

Simulated world

- Real terrain, 4 areas
- 4 drone types
- Real flight limits, wind

Measured world

- Radio links per second
- Diffraction and fading
- Fixes with their error

Export

- truth.csv answer key
- TAK Cursor on Target
- Same seed, same bytes

NEXT

Advanced simulation
wind and weather by height, land cover, jamming

More detector types
radar, camera, thermal, acoustic, Remote ID

Action recommendations
where to place the next sensor, how to respond

LIVE DEMO · 60 SECONDS

Plan · Simulate · Export

1 **Plan: pick an area, place receivers, draw drone paths**

set the drone mix, wind and mast height, then run

`localhost:8765/planner/`

2 **Simulate: watch the receivers at work**

bearings, fixes with their error, the coverage heatmap in 3D

`localhost:8765/viewer/`

3 **Export: the files behind it**

truth.csv · rf_raw.csv · fused.cot.xml, ready for TAK

`out/planned/ · out/<area>/`